Scientific process and research workflow

Workflow labs use the variant template: Goal → Approach → Execution → Check → Report.

Learning objectives

- Name the nine stages of the research workflow and say what goes wrong if each is skipped.
- Distinguish biological from statistical significance with an example.
- Frame a research question that can survive a methods section.

Prerequisites

R, RStudio, and Quarto installed (see Get started).

Background

Statistics is a thread running through a longer process that begins before a dataset exists. The research workflow — Question → Measurements → Design → Acquisition → Description → Analysis → Interpretation → Validation → Knowledge — names the stages of that process and makes clear that errors made at one stage are paid for at the next. A well-posed question protects you from a bad design; a good design protects you from a noisy dataset; a clean dataset protects you from needing exotic methods. Much of what looks like statistical failure in the literature is in fact a workflow failure three stages earlier.

The workflow also organises how you write up the work. The Introduction names the question; the Methods describe the measurements, design, and analysis; the Results describe the data and the inference; the Discussion interprets. When you plan an analysis in this order, the paper nearly writes itself. When you skip a stage, reviewers notice exactly which one.

Setup

```r
library(tidyverse)
set.seed(42)
theme_set(theme_minimal(base_size = 12))
```

1. Goal

Make the research workflow concrete by simulating a small clinical scenario and mapping each stage to a specific decision.

2. Approach

Imagine a new blood-pressure drug. Our simulated question: *does drug X reduce systolic blood pressure compared with placebo in adults with hypertension?* We will generate data that reflect a realistic effect and walk through the workflow.

```r
df <- tibble(
  id    = seq_len(n),
  arm   = rep(c("placebo", "drug"), each = n / 2),
  sbp_0 = rnorm(n, mean = 150, sd = 10),
```

```r
  sbp_1 = sbp_0 + ifelse(arm == "drug", -6, 0) + rnorm(n, 0, 8)
)
```

3. Execution

```r
df |>
  mutate(delta = sbp_1 - sbp_0) |>
  ggplot(aes(arm, delta, fill = arm)) +
  geom_boxplot(alpha = 0.6, colour = "grey30") +
  labs(x = NULL, y = "Change in SBP (mmHg)") +
  theme(legend.position = "none")
```

The plot is the Description stage. Without it, we would skip straight from raw numbers to a test. With it, the direction and spread of the change are visible before a single calculation.

4. Check

```r
fit
```

The drug arm has a lower change-in-SBP on average; the interval is compatible with a clinically meaningful effect.

5. Report

> In a simulated two-arm trial ($n = 120$), mean change in systolic blood pressure was `round(diff(tapply(df$sbp_1 - df$sbp_0, df$arm, mean)), 1)` mmHg lower in the drug arm than in placebo (95% CI: `round(fit$conf.int[1], 1)` to `round(fit$conf.int[2], 1)`; p = `signif(fit$p.value, 2)`).

Note the three pieces every report needs: direction and magnitude, an interval, and a p-value reported alongside — never instead of — the effect.

Biological vs statistical significance

A p-value below 0.05 does not, on its own, mean the effect matters clinically. A reduction of 0.1 mmHg in SBP would be statistically significant with a million patients but biologically meaningless. The discipline of asking *how large is the effect?* — not *is there any effect?* — is the single most important habit to form early on.

Common pitfalls

- Skipping the description stage and running a test on data you have never plotted.
- Reporting a p-value without an effect size and interval.
- Conflating biological and statistical significance.
- Treating the research workflow as optional once data are in hand.

Further reading

- Research workflow appendix.
- Altman DG. *Practical Statistics for Medical Research*, ch. 1.
- Wasserstein RL & Lazar NA (2016), *The ASA's Statement on p-Values*.

Session info

See also — chapter index

- Methods in this chapter
- Find your method (Chapter 0)